

SCOPE: Algorithms for each causal learner

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1 S-learner

Algorithm 1 SCOPE-S: Training and Inference (Case-aware)

Training (Backward Pass; Offline)

Input: Dataset $\mathcal{D} = \{(\sigma_{:l_k}^{(c)}, a_k^{obs, (c)}, y^{(c)})\}_{c=1, k=1}^{C, K}$, action spaces $\{\mathcal{A}_k\}_{k=1}^K$

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1:  $\hat{V}_{K+1}^{(c)} = y^{(c)}$  for all  $c$  ▷ Initialize value for each case
2:  $\mathbf{M} = []$ 
3: for  $k = K, K-1, \dots, 1$  do ▷ Iterate backwards
4:    $\mathcal{M}_k \leftarrow \text{Regress } \hat{V}_{k+1}^{(c)} \text{ on } (\sigma_{:l_k}^{(c)}, a_k^{obs, (c)}) \text{ on all } c$  ▷ S-learner to estimate Q-function
5:   Append  $\mathcal{M}_k$  to  $\mathbf{M}$ 
6:   for each case  $c = 1, \dots, C$  do
7:      $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in \mathcal{A}_k} \mathcal{M}_k(\sigma_{:l_k}^{(c)}, a_k)$  ▷ Estimated optimal action
8:      $\hat{Q}_k^{obs, (c)} = \mathcal{M}_k(\sigma_{:l_k}^{(c)}, a_k^{obs, (c)})$  ▷ Q-value of observed action
9:      $\hat{Q}_k^{opt, (c)} = \mathcal{M}_k(\sigma_{:l_k}^{(c)}, \hat{a}_k^{opt, (c)})$  ▷ Q-value of optimal action
10:     $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)} + \hat{Q}_k^{opt, (c)} - \hat{Q}_k^{obs, (c)}$  ▷ Compute value via regret-based value function
11:  end for
12: end for
13: return  $\mathbf{M}$ 

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Inference (Forward Pass; Online)

Input: Models $\{\mathcal{M}_k\}_{k=1}^K$, $\sigma_{:l_1}^{(c)}$ for each c

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14: for  $k = 1, 2, \dots, K$  do ▷ Iterate forwards
15:   for each case  $c = 1, \dots, C$  do
16:     Get  $\sigma_{:l_k}^{(c)}$  (given previous action if  $k > 1$ )
17:      $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in \mathcal{A}_k} \mathcal{M}_k(\sigma_{:l_k}^{(c)}, a_k)$  ▷ Estimated optimal action
18:   end for
19: end for

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2 T-learner

Algorithm 2 SCOPE-T: Training and Inference (Case-aware)

Training (Backward Pass; Offline)
Input: Dataset $\mathcal{D} = \{(\sigma_{:l_k}^{(c)}, a_k^{obs,(c)}, y^{(c)})\}_{c=1, k=1}^{C, K}$, action spaces $\{\mathcal{A}_k\}_{k=1}^K$

- 1: $\hat{V}_{K+1}^{(c)} = y^{(c)}$ for all c ▷ Initialize value for each case
- 2: $\mathbf{M} = []$
- 3: **for** $k = K, K-1, \dots, 1$ **do** ▷ Iterate backwards
- 4: $\mathcal{M}_k = []$ ▷ Fit a separate model for each action
- 5: **for** each action $a_k \in \mathcal{A}_k$ **do**
- 6: $\mathcal{M}_k[a_k] \leftarrow$ Regress $\hat{V}_{k+1}^{(c)}$ on $\sigma_{:l_k}^{(c)}$ using only cases with $a_k^{obs,(c)} = a_k$ ▷ Fit a T-learner
- 7: **end for**
- 8: Append $\mathcal{M}_k$ to $\mathbf{M}$
- 9: **for** each case $c = 1, \dots, C$ **do**
- 10: $\hat{a}_k^{opt,(c)} = \operatorname{argmax}_{a_k \in \mathcal{A}_k} \mathcal{M}_k[a_k](\sigma_{:l_k}^{(c)})$ ▷ Estimated optimal action
- 11: $\hat{Q}_k^{obs,(c)} = \mathcal{M}_k[a_k^{obs,(c)}](\sigma_{:l_k}^{(c)})$ ▷ Q-value of observed action
- 12: $\hat{Q}_k^{opt,(c)} = \mathcal{M}_k[\hat{a}_k^{opt,(c)}](\sigma_{:l_k}^{(c)})$ ▷ Q-value of optimal action
- 13: $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)} + \hat{Q}_k^{opt,(c)} - \hat{Q}_k^{obs,(c)}$ ▷ Compute value via regret-based value function
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{M}$

Inference (Forward Pass; Online)
Input: Model lists $\{\mathcal{M}_k\}_{k=1}^K$, $\sigma_{:l_1}^{(c)}$ for each c

- 17: **for** $k = 1, 2, \dots, K$ **do** ▷ Iterate forwards
- 18: **for** each case $c = 1, \dots, C$ **do**
- 19: Get $\sigma_{:l_k}^{(c)}$ (given previous action if $k > 1$)
- 20: $\hat{a}_k^{opt,(c)} = \operatorname{argmax}_{a_k \in \mathcal{A}_k} \mathcal{M}_k[a_k](\sigma_{:l_k}^{(c)})$ ▷ Estimated optimal action
- 21: **end for**
- 22: **end for**

3 RA-learner

Algorithm 3 SCOPE-RA: Training and Inference (Case-aware)

Training (Backward Pass; Offline)

Input: Dataset $\mathcal{D} = \{(\sigma_{:l_k}^{(c)}, a_k^{obs, (c)}, y^{(c)})\}_{c=1, k=1}^{C, K}$, action spaces $\{\mathcal{A}_k\}_{k=1}^K$

- 1: $\hat{V}_{K+1}^{(c)} = y^{(c)}$ for all c ▷ Initialize value for each case
- 2: $\mathbf{M} = []$
- 3: **for** $k = K, K-1, \dots, 1$ **do** ▷ Iterate backwards
- 4: $\mathcal{M}_k \leftarrow$ Regress $\hat{V}_{k+1}^{(c)}$ on $(\sigma_{:l_k}^{(c)}, a_k^{obs, (c)})$ on all c ▷ S-learner to estimate Q-function
- 5: Append $\mathcal{M}_k$ to $\mathbf{M}$
- 6: **for** each case $c = 1, \dots, C$ **do**
- 7: **for** each action $a_k \in \mathcal{A}_k$ **do**
- 8: $\Phi_k^{(a_k)(c)} \leftarrow$ Compute pseudo-outcome using $(\sigma_{:l_k}^{(c)}, \mathcal{M}_k, a_k)$
- 9: **end for**
- 10: **end for**
- 11: $\mathcal{E}_k \leftarrow$ Regress $\Phi_k^{(a_k)(c)}$ on $(\sigma_{:l_k}^{(c)}, a_k)$ on all c ▷ Second model to estimate causal effects
- 12: **for** each case $c = 1, \dots, C$ **do**
- 13: $\hat{a}_k^{opt, (c)} = \operatorname{argmax}_{a_k \in \mathcal{A}_k} \mathcal{E}_k(\sigma_{:l_k}^{(c)}, a_k)$ ▷ Estimated optimal action
- 14: $\hat{Q}_k^{obs, (c)} = \mathcal{M}_k(\sigma_{:l_k}^{(c)}, a_k^{obs, (c)})$ ▷ Q-value of observed action
- 15: $\hat{Q}_k^{opt, (c)} = \mathcal{M}_k(\sigma_{:l_k}^{(c)}, \hat{a}_k^{opt, (c)})$ ▷ Q-value of optimal action
- 16: $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)} + \hat{Q}_k^{opt, (c)} - \hat{Q}_k^{obs, (c)}$ ▷ Compute value via regret-based value function
- 17: **end for**
- 18: **end for**
- 19: **return** $\mathbf{M}$

Inference (Forward Pass; Online)

Input: Causal effect models $\{\mathcal{E}_k\}_{k=1}^K, \sigma_{:l_1}^{(c)}$ for each c

- 20: **for** $k = 1, 2, \dots, K$ **do** ▷ Iterate forwards
- 21: **for** each case $c = 1, \dots, C$ **do**
- 22: Get $\sigma_{:l_k}^{(c)}$ (given previous action if $k > 1$)
- 23: $\hat{a}_k^{opt, (c)} = \operatorname{argmax}_{a_k \in \mathcal{A}_k} \mathcal{E}_k(\sigma_{:l_k}^{(c)}, a_k)$ ▷ Estimated optimal action
- 24: **end for**
- 25: **end for**
